



Global boundaries, task processes and IS project success: a field study

IS project
success: a field
study

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Abstract

Purpose – The purpose of the paper is to better understand how global boundaries affect global information system (IS) project success and which mediating process variables increase the chance of success.

Design/methodology/approach – Based on the literature on IS success and global teams, an input-process-output framework is adopted to develop the research model for the study. This research is based on semi-structured interviews with 22 global IS project managers. An attribution analysis is used to identify common themes and patterns of the interview results.

Findings – Global IS project managers identified time separation and cultural differences as the most significant barriers to project success. Our findings suggest that effective teams were able to overcome these barriers to achieve success, but this success was achieved through the implementation of special coordination, communication and cognitive processes tailored to help teams overcome global barriers and through considerable additional cost and effort.

Practical implications – This study furthers understanding of the global boundaries affecting global IS project success and the most effective processes that teams use to overcome global barriers.

Originality/value – Despite the increasing attention to global IS work, there is limited understanding of why and how global IS projects succeed or fail. The present study, investigates not only how multiple global boundary variables (e.g. geographic dispersion, time separation, language differences, cultural differences, etc.) affect IS project success, but also which processes teams use to cope with the challenges presented by these global boundaries.

Keywords Information systems, Project management, Project teams, Team working

Paper type Research paper

Introduction

Information system (IS) projects increasingly involve global teams as organizations seek to make the most of internal and external resources that are globally dispersed. IS project teams often consist not only of domestic employees, but also of employees and outsourcing personnel from other countries. While such “global” work contexts often afford proximity to clients and skilled personnel, they make it more difficult for team members to communicate task information and coordinate task execution and increase project complexity, which can undermine IS project performance (Xia and Lee, 2004, 2005). Therefore, one of the key challenges for IS organizations today is to find ways to make their global project teams work effectively and to deliver quality systems on time and within budget. Despite the increasing attention to global IS work, there is limited



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knowledge of why and how global IS projects succeed or fail, partly because of the difficulty in understanding how various global barriers jointly affect IS project success.

Global teams working on IS (Orlikowski, 2002) and other projects (Drucker, 1993; DeSanctis and Poole, 1997) are often divided by multiple boundaries, including time zones, geographic distance, and cultural differences, which present substantial challenges for team members who need to bridge the “discontinuities” caused by the presence of these boundaries (Chudoba *et al.*, 2002). Past studies have mainly focused on just one or two boundaries, often ignoring other important boundaries (Watson-Manheim *et al.*, 2002a). Because global boundary variables often co-vary, the respective effects of a given global boundary variable on IS project success can be confounded if the effects of other global boundary variables are not accounted for (Espinosa *et al.*, 2003). In the present study we attempt to fill this gap by empirically investigating the simultaneous effect of these various global team boundaries on various dimensions of IS project success and by identifying the most effective processes for bridging those boundaries. We draw on prior research on global teams, virtual teams and IS success to formulate and empirically validate our framework for global IS project success, which extends prior frameworks suggested for virtual teams research (Powell *et al.*, 2004) by including additional processes specifically implemented to effectively bridge global boundaries. This study is exploratory and qualitative in nature and therefore cannot completely tease out all the various effects. Nevertheless it represents the first attempt to empirically investigate which global boundary variables have the strongest effect on IS project success when considered jointly. Our findings contribute to research and practice by helping us understand how dependencies across global boundaries can be managed more effectively.

Our two research questions are, therefore:

- RQ1. When considered jointly, which global boundary variables have the strongest effect on global IS project success, and how?
- RQ2. Which mediating task process variables increase the chance of success, given the presence of these boundary variables, and how?

This study investigates these questions empirically based on 22 interviews with global IS project managers. In the following section we present the theoretical foundations of our study and propose a research framework. Subsequent sections include research methodology, results, discussion and concluding remarks.

Theoretical framework

The research framework for this study follows the “input-process-outcome” (I-P-O) model widely used to study group effectiveness (McGrath, 1984; Hackman, 1987). Powell and colleagues (1996) used an I-P-O framework, illustrated in Figure 1, to

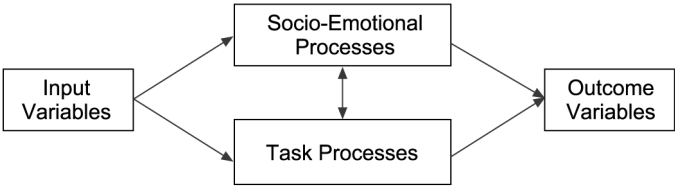


Figure 1.
Powell et al. virtual team
research framework

conduct an extensive literature review on global teams and further categorized process variables into socio-emotional and task processes. The research framework for the present study is based on Powell *et al.*'s I-P-O framework.

The inputs in the I-P-O model in the present study are global boundaries and other project context variables. With respect to process variables, our study focuses on task processes because we are interested in learning what global teams do to accomplish their IS project goals. We build upon and extend the Powell *et al.* framework by: distinguishing between general task process that traditional teams use to carry out their teamwork and coping processes utilized to bridge global boundaries; focusing more specifically on the latter; including cognitive process variables. The outcome variables represent IS project success variables and other outcomes. The extended framework adopted by this study is further discussed and illustrated in the following sections.

Outcomes: global IS project success

The research literature suggests that IS project performance consists of two different dimensions: process performance and product performance (Cooprider and Henderson, 1991; Nidumolu, 1995; Wixom and Watson, 2001). Process performance refers to how well the IS project process has been undertaken. It is measured by on-time/on-budget completion of the project, user participation, team member satisfaction and morale, etc. (Powers and Dickson, 1973; Nidumolu, 1995; Deephouse *et al.*, 1996). Product performance refers to the performance of the information system, including system quality, system functionality, system impact, and user satisfaction about the system. Based on these studies and the DeLone and McLean's (2003) updated IS Success Model, this study includes the following measures of IS project success: on-time completion, within-budget completion, system costs/effort, meeting system requirements, system quality, user satisfaction, project team satisfaction, system use, and net system benefits.

Inputs: global team boundary variables

The main input variables of interest for our study are global team boundaries. Research with software organizations (Orlikowski, 2002), published articles on virtual teams (Watson-Manheim *et al.*, 2002b), and case studies (Espinosa *et al.*, 2003) have identified boundaries that global team members need to bridge to work together, including geographic distance, time separation, organizational, functional, and cultural. As more of these boundaries are present, the "virtuality" of the team increases (Lu *et al.*, 2006), creating "fault lines" (Lau and Murnighan, 1998) making it more difficult for team members to work together.

Geographic distance. A number of research studies have examined the effects of distance on teamwork (Conrath, 1973; Allen, 1977; Van den Bulte and Moenaert, 1998). Geographic separation is generally associated with: increased coordination challenges, more delays, communication problems, and differences in feedback cycles (Carmel, 1999; Herbsleb *et al.*, 2000); misunderstandings and conflict escalation (Armstrong and Cole, 2002); reduced informal or spontaneous communication and social cohesion (Kiesler and Cummings, 2002); reduced interaction and shared understanding of the task context (Malhotra *et al.*, 2001); less familiarity about the task and team members

(Hinds and Bailey, 2003); and lower levels of collective knowledge (Griffith *et al.*, 2003). These factors affect the team members' ability to coordinate their task activities.

Time separation. Global collaborators are often separated by time due to differences in working hours, time zones, and/or working cycles. This reduces the time available for synchronous interaction (Espinosa *et al.*, 2003; Espinosa and Carmel, 2004), particularly when team members are scattered from east to west (O'Leary and Cummings, 2002). Such time separation makes it difficult to coordinate and communicate, making teams more "virtual" (Griffith *et al.*, 2003), particularly as more time zones are represented (Espinosa and Pickering, 2006).

Organizational differences. The increasing popularity of global outsourcing (Carmel, 2005) is leading to an increase in the utilization of teams that cross organizational boundaries (Yan and Louis, 1999; Majchrzak *et al.*, 2000). Research on distributed work groups often examines teams representing multiple organizations (Robey *et al.*, 2000; Cramton, 2001; Armstrong and Cole, 2002). Differences in organizational affiliations can: reduce shared understanding of context, inhibit a group's ability to develop a shared sense of identity, and affect communication and performance effectiveness (Zack and McKenney, 1995); and increase hidden costs (Barthelemy, 2001) because client organizations need to communicate work requirements to information technology (IT) service providers and monitor contractual obligations.

Functional diversity. Some problems in global work have to do with functional differences that may exist between sites (Olson and Olson, 2000) when more than one area of functional expertise is represented within a team (Ancona and Caldwell, 1992; Eisenhardt and Tabrizi, 1995; Denison *et al.*, 1996). Functional differences can influence team processes, which in turn can affect the group's performance (Jehn and Northcraft, 1999; Pelled *et al.*, 1999).

Cultural differences. Cultural differences can make collaboration and communication more difficult (Carmel, 1999; Orlikowski, 2002; Watson-Manheim *et al.*, 2002b). Studies of global software teams have found that distance alone might not be the only cause of observed problems, but that cultural and other differences also can play a role (Cramton, 2001; McDonough *et al.*, 2001), and that indeed some of the problems of geographic dispersion have to do with cultural differences when work is done globally (Olson and Olson, 2000).

Language differences. Global team members also need to bridge language differences within the team, which can be an impediment for global collaboration, particularly when there is ambiguity and lack of visual channels in the task (Olson and Olson, 2000).

Other input variables. Input variables other than global boundaries that which may affect the effectiveness of global collaboration and project outcomes include: prior global work experience, prior work experience with team members (Goodman and Leyden, 1991) and task context and dependency (McGrath, 1984).

Task process variables

A review of the literature identified two main categories of task process variables that influence performance. The first category is task coordination, which is important in IS projects (Curtis *et al.*, 1988; Kraut and Streeter, 1995; Crowston and Kammerer, 1998), particularly when the work is carried out globally (Herbsleb and Grinter, 1999;

Espinosa *et al.*, 2002, 2003). The second category includes cognitive variables, which are important for global team work because they help members work more effectively with reduced communication because members can make “unspoken assumptions about what others in the group are likely to do” (Wittenbaum and Stasser, 1996).

Coordination: task programming and team communication. Coordination involves managing dependencies among task activities (Malone and Crowston, 1990, 1994), which is a difficult thing to do across multiple global boundaries. Team members use two types of coordination mechanisms: task programming and team communication (March and Simon, 1958; Thompson, 1967; Van de Ven *et al.*, 1976). Coordination of repetitive and routine aspects of the task can be “programmed” using mechanisms like tools, schedules, plans, division of labor, project controls and specifications. But more uncertain or less routine aspects of the task can be more effectively coordinated through communication, which can be synchronous (e.g. face-to-face, telephone, instant messaging) or asynchronous (e.g. electronic mail, shared databases). When collaborators are in close proximity they often meet spontaneously and coordinate their work informally (Perry *et al.*, 1994; Kraut and Streeter, 1995; Teasley *et al.*, 2002). As global barriers impair communication among members, task programming mechanisms can help members coordinate with reduced communication.

Team cognition. Team cognition can be a great complement to traditional coordination mechanisms because it helps teams coordinate implicitly with reduced communication. This study explores three types of team cognition: shared knowledge, shared beliefs and trust. Studies have shown that shared knowledge has a positive effect on IS project outcomes (Nelson and Coopridge, 1996). It provides a common ground for effective communication (Cramton, 2001) with less complex messages (Maznevski and Chudoba, 2000) and a common knowledge base that helps team members tap into expert knowledge sources within the team (Alavi and Leidner, 2001). Beliefs that team members share about things like goals, strategies, individual capabilities, and task priorities are also important because members have a shared understanding of the group’s task and each other, which helps members plan their individual actions more effectively (Cannon-Bowers *et al.*, 1993; Klimoski and Mohammed, 1994; Rentsch and Hall, 1994; Cooke *et al.*, 2000; Mathieu *et al.*, 2000), especially with IS project teams (Crowston and Kammerer, 1998).

Trust is important for global IS project success because it can reduce transaction costs (Cummings and Bromiley, 1996) and facilitate information exchange (Earley, 1986). Trust lowers transaction costs of relationships because individuals engage less in self-protective actions (Kramer and Tyler, 1996). Trust is also critical in preventing geographical distance from leading to psychological distance within a global team (Snow *et al.*, 1996). However, global boundaries like geographic distance and cultural differences may make it difficult to develop trust (O’Hara-Devereaux and Johansen, 1994; Handy, 1995; Lee and Kim, 1999). Based on the theoretical foundations discussed above and summarized in Table I, the research framework and model for this study are represented in Figures 2 and 3.

Research methodology

Research context and sample

We conducted semi-structured interviews face-to-face and by telephone with managers of global IS projects. Interviews are a widely used method in information systems

Global team boundaries	<i>Input variables</i>	
	(Lau and Murnighan, 1998; Orlikowski, 2002; Watson-Manheim <i>et al.</i> , 2002b; Espinosa <i>et al.</i> , 2003; Lu <i>et al.</i> , 2003)	
	Geographic distance	
	Conrath, 1973; Allen, 1977; Van den Bulte and Moenaert, 1998; Carmel, 1999; Herbsleb <i>et al.</i> , 2000; Malhotra <i>et al.</i> , 2001; Armstrong and Cole, 2002; Kiesler and Cummings, 2002; Griffith <i>et al.</i> , 2003; Hinds and Bailey, 2003)	
	Time separation	
	(O'Leary and Cummings, 2002; Espinosa <i>et al.</i> , 2003; Griffith <i>et al.</i> , 2003; Espinosa and Carmel, 2004; Espinosa and Pickering, 2006)	
	Organizational differences	
	(Zack and McKenney, 1995; Yan and Louis, 1999; Majchrzak <i>et al.</i> , 2000; Robey <i>et al.</i> , 2000; Barthelemy, 2001; Cramton, 2001; Armstrong and Cole, 2002)	
	Functional diversity	
	(Ancona and Caldwell, 1992; Eisenhardt and Tabrizi, 1995; Denison <i>et al.</i> , 1996; Jehn and Northcraft, 1999; Pelled <i>et al.</i> , 1999; Olson and Olson, 2000)	
	Cultural differences	
	(Carmel, 1999; Olson and Olson, 2000; Cramton, 2001; McDonough <i>et al.</i> , 2001; Orlikowski, 2002; Watson-Manheim <i>et al.</i> , 2002b).	
	<i>Coping process variables</i>	
	Coordination	
	(Curtis <i>et al.</i> , 1988; Malone and Crowston, 1990, 1994; Kraut and Streeter, 1995; Crowston and Kammerer, 1998; Herbsleb and Grinter, 1999; Espinosa <i>et al.</i> , 2002, 2003)	
Communication Task programming	Communication	
	(March and Simon, 1958; Thompson, 1967; Van de Ven <i>et al.</i> , 1976; Allen, 1977; Perry <i>et al.</i> , 1994; Kraut and Streeter, 1995; Teasley <i>et al.</i> , 2002)	
	Team cognition	
	(Wittenbaum and Stasser, 1996)	
	Shared knowledge	
	(Nelson and Coopridge, 1996; Alavi and Leidner, 2001; Cramton, 2001)	
	Shared beliefs	
	(Cannon-Bowers <i>et al.</i> , 1993; Klimoski and Mohammed, 1994; Rentsch and Hall, 1994; Crowston and Kammerer, 1998; Cooke <i>et al.</i> , 2000; Mathieu <i>et al.</i> , 2000)	
	Trust	
	(Earley, 1986; Jarvenpaa and Ives, 1994; O'Hara-Devereaux and Johansen, 1994; Handy, 1995; Cummings and Bromiley, 1996; Kramer and Tyler, 1996; Snow <i>et al.</i> , 1996; Jarvenpaa <i>et al.</i> , 1998; Jarvenpaa and Leidner, 1999; Lee and Kim, 1999)	
IS project success	<i>Outcome variables</i>	
	IS project success	
	(Coopridge and Henderson, 1991; Nidumolu, 1995; Wixom and Watson, 2001; DeLone and McLean, 2003)	
	On time;	
	On budget;	
	System quality;	
	Cost/effort meet requirements;	
	User satisfaction;	
	Team satisfaction;	
	System use;	
	Net system benefits	

Table I.
Literature support for
research model for global
IS project success

Note: the numbers indicate the number of interviews in which the success measure was mentioned

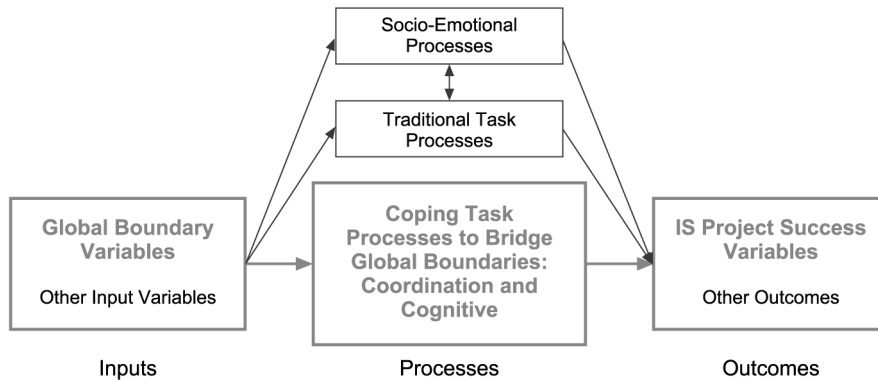


Figure 2.
Research framework for
this study

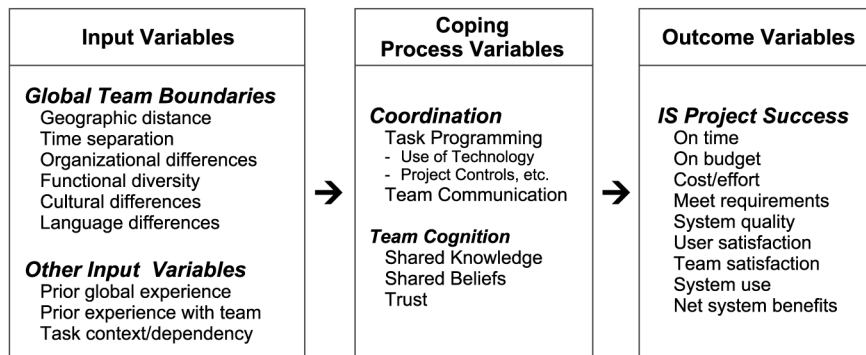


Figure 3.
Research model for IS
project success with
variables of interest

research for exploratory studies (Orlikowski, 1993; Majchrzak *et al.*, 2000; Malhotra *et al.*, 2001; Orlikowski, 2002). Interview questions focused on the relationships between global boundaries and project performance, and also inquired about the processes employed to mitigate the potential negative impacts of global boundaries. We used the criterion of “theoretical saturation” recommended by grounded theory to set the sample size, which recommends concluding the interview process when the last few interviews do not contribute new insights to the inquiry (Strauss and Corbin, 1998). This criterion yielded 22 interviews of global IS project managers from seven different organizations located in Australia, India, Ireland, Mexico, South Africa, the UK and the USA (east coast and west coast).

The seven organizations in our study sample represent the automotive, music, computer, financial, and IT services industries. All participants had significant responsibilities for their projects, ranging from development to project management. Six participants were IT executives and another 16 participants were managers. On average, they had 6.6 years of work experience managing global software projects. Most of the software projects in our sample were application development or enhancement projects; 14 projects involved one or more outsourcing partners while the other projects were globally insourced. The projects involved from two to 12 development sites; 13 projects reported project budgets ranging from \$500,000 to \$45

million, with the mean of \$11 million, and two additional projects reported scope in terms of 6,100 and 13,600 person-hours. Project duration ranged from six months to five years, with the mean of 2.2 years. Projects involved from one to 28 corporate divisions, with a median of four divisions. In summary, the sample projects were global and large in scale.

Data collection

Interviews averaged 45 minutes and were audio taped with permission from the participant. Participants were first asked a few background questions and then were asked to define how they measured success of IS projects (i.e. the dependent variable). Participants were then asked questions intended to generate attributions about the general effect of global boundary variables on success outcomes (I(O) without tying these answers to specific projects. Finally, participants were asked to identify a recent important global IS project that was salient in their minds, and all remaining questions were asked in reference to that specific project. These questions were intended to generate attributions about which task processes were employed to work across global boundary variables (I(P) and how these processes impacted project success (P(O). The interview instrument can be found in the Appendix.

Coding and data analysis

The interviews were transcribed verbatim into 283 single-spaced pages of text with 117,478 words in total. The text was first analyzed and open coded to discover recurrent themes around attributions and then axial coded to find relationships among themes (Strauss and Corbin, 1998) in order to develop a template with hierarchical codes, which we then used for codebook analysis (King, 1998). The hierarchical codes contained three primary high-order codes (i.e. I: input variable; P: process variable; and O: outcome variable). The sub-codes in the hierarchy identified the specific type of variable associated with the high-order variable. For example, P.Cm.S is a code that identified synchronous (S) communication (Cm), which is a process variable (P). We also included a valence dimension on outcome variables to identify the direction of the success outcome as P, N and 0 for positive, negative and no effect, respectively. The interview text was then coded one final time to identify general attributions (Silvester, 1998) of the effect of one type of variable over another. For example, a code of "I.D → O.T-N" identifies a statement in which the input variable (I) geographic distance (D) was attributed as affecting the success outcome variable (O) on-time delivery (T), and that this effect was negative (N).

The first few interviews were coded independently by each of the researchers and all coding differences were discussed and resolved by consensus, after which the remaining interviews were coded, resulting in 703 unique coded text segments. An external coder was trained on the coding scheme and then instructed to code a few cases. After analyzing all discrepancies with the external coder, these were resolved by mutual agreement and the external coder re-coded this initial set of interviews and then coded all remaining interviews. The final reliability between the researchers' consensus coding and the external coder was 71.5 percent, measured as the number of agreements over agreements plus disagreements (Miles and Huberman, 1994), indicating an acceptable level of reliability (Landis and Koch, 1977; Miles and Huberman, 1994).

In total, 683 of the coded text segments contained $I \rightarrow O$, $I \rightarrow P$ or $P \rightarrow O$ attributions, or an average of 31.0 attributions per interview. Of these, 245 were $I \rightarrow O$ attributions or 11.1 per interview; 274 were $I \rightarrow P$ attributions or 12.4 per interview; and 164 were $P \rightarrow O$ attributions or 7.8 per interview. We then analyzed the text data by sorting all coded segments by their respective cause and effect attribution variables and analyzing all similar attributions made in an interpretative manner to identify cause-effect trends. All trends identified were jointly discussed, evaluated and agreed upon by all researchers.

Results

Outcomes: global IS project success

Despite the risks inherent in multiple global boundaries, participants rated 21 of the 22 projects as successful based on a variety of global IS success measures. However, several project managers indicated that budgets and timelines had to be revised at least once during a project to account for the additional effort needed to make global projects succeed in the presence of global barriers. One manager estimated that budgeted resources and timelines for global projects were 1.5 times larger and longer than budgeted resources and timelines for same size domestic projects.

Attributions involving global boundary and other input variables

Table II summarizes attributions about the impacts of input variables on project outcomes, indicating the total number of unique attributions made by participants and the total number of projects in which these attributions were made. Note that the “General Outcome” category refers to generic success attributions that did not identify a specific success measure such as on-budget. We now discuss these attributions starting with the most frequently cited variables.

Time separation. Discussed by 17 or 77 percent of the participants, time separation was by far the most frequently cited global boundary variable affecting project outcomes. Time separation is an interesting global variable in that it can be used advantageously because team members can advance project work during other members’ off-work hours. Eight participants discussed the beneficial effects of time separation; five of them specifically mentioned its positive effect on on-time delivery; and two others mentioned that it actually reduced effort. However, time separation can be detrimental when the timing of the work can’t be programmed with certainty, when there are tight deadlines or when the team needs to interact frequently. In total, 13 participants stated that time separation had a negative effect on project outcomes; eight mentioned that it had a detrimental effect on on-time delivery; and four others stated that it required more effort.

A number of factors affect the difficulty of working across time zones, including: whether time differences are aligned with workflows (i.e. work is handed over at the end of the work day to a site whose work day is just beginning); the number of different time zones represented within a team; and the magnitude of the time difference. As one participant commented, “we only had a two-hour time window to work together, which is difficult . . . a task that should take two or three weeks might take a month or two.”

An important consequence of time separation is that collaborators need to make conscious choices about the timing and mode of interaction (i.e. communicate asynchronously during non-overlapping hours or wait until overlapping hours to

Table II.

Input variable	General outcome	On budget	Costs and effort	Syst quality	Outcome On time	Meet reqs	Subtotal unique attributions	Total projects
Global boundary variables								
Geographic distance	8		5	1	3	1	18	11
Time separation	16		11		11		38	17
Organizational differences	6	1	1	1	2		11	7
Functional diversity	3			1	1		5	2
Cultural differences	12	1	2	2	3		20	13
Language differences	8		2		1		11	8
<i>Other input variables</i>								
Prior global experience	9	1	7		1		18	10
Prior experience w/team	1		1	2	1		5	5
Task								
context/dependencies	3	1	1	1	3		9	6
Total	66	4	30	8	26	1	135	79

Note: the numbers indicate the number of interviews in which the attribution was made

communicate synchronously). Time differences bring about a substantial burden in the time demands and personal lives of team members. One participant commented that the way they addressed problems of time difference was by being available 24 hours a day to their team members through pagers and mobile phones, often having to communicate from their homes. The benefits of spontaneous team interaction within teams without such practices are severely hampered with time separation.

Cultural differences. Cultural differences were attributed to project outcomes by 13 participants, most of whom discussed negative effects on project performance in terms of time overruns, budget overruns, higher cost/effort, and lower system quality. Project team members from different countries often had different perspectives on issues and different ways of communicating and resolving them. Team members often experienced difficulties in understanding and predicting other members' behavior and working as one team. Subtle cognitive differences due to cultural gaps require more time and effort to resolve. "It takes a lot longer time to figure out certain things that would probably get resolved by a five minute phone call . . . they dwell on the issue or problem for probably a week or two weeks," said a participant. In particular, differences between Asian culture and Western culture often created confusion and hindered project success. For example, Asian team members hesitated to say no to others or to openly bring up potential problems, as one participant commented: "they're very reluctant to let you know if something is not going to schedule . . . they feel that if they're telling you a problem, it reflects badly on them."

The negative effect of cultural differences on project performance seemed especially salient in the initial phase of projects, making it important to take early measures to address cultural differences within teams. Early awareness of cultural differences helped project team members mitigate potential problems and risks. As people accumulate work experience with people from different cultures, they become better equipped to cope with the challenge.

Language differences. Language differences caused time overruns, extra cost and effort, and lower system quality in eight projects. In virtually all projects, English was used as a common business language when project team members spoke different languages. However, even when all team members spoke a common business language, various problems emerged. Non-native speakers usually had difficulties in reading between the lines and understanding subtle differences in what is being communicated. As a result, language barriers caused reduced project participation of non-native speakers, less frequent communications, longer time for communications, and misunderstandings, which eventually led to lower project performance, as this participant suggested: "when you're dealing with people who are not native speakers of English, then suddenly the ability to really get across nuances of what's really going on becomes an issue."

Geographic distance. Distance was attributed to project outcomes in 11 projects. Interestingly, contrary to the conventional belief, geographic distance was not a major impediment according to half of the participants. Comments ranged from "I didn't see it [distance] as a problem" to "[distance is] part of the reason the project wasn't working so well in that everybody was working remotely". However, reducing or eliminating the impact of geographic dispersion was costly. For example, five participants stated that they had to invest more resources to overcome the difficulties of working over geographic distance. As one participant explained "[the project] required frequent

travel and one of the impacts was that we had a significant budget for travel in the project financials". Another participant characterized such frequent travel as "reducing the psychological distance" among team members because familiarity with teammates and their contexts reduces this "perceived" distance.

Organizational differences. Seven of the 14 projects with outsourcing arrangements stated that organizational boundaries affected project outcomes. Interestingly, the impacts were found to be both positive and negative. In some cases, working with multiple external organizations was viewed as an advantage because multiple companies may reduce project risks, increase and diversify skill sets and create competition that leads to better project performance. On the other hand, managing multiple vendors increases project complexity as a result of different and sometimes conflicting sets of goals, agreements, responsibilities and success measures. As one participant commented "[vendors] ... have their own vested interest, or the way they look at the project's success may be different from somebody trying to manage the overall global project."

Prior experience with team. Prior experience working with the current team members had a positive effect on global project success outcomes according to five participants, as this comment illustrates: "sure, it [prior work experience with the team] makes a lot of difference because everybody understands each other, the way they work, and what is the performance level of each individual, and you have worked as a team ... It will surely give a positive impact." Two participants indicated that their team did not have such experience working together, so they had to spend some time building team relations at the beginning of the project.

Prior global project experience. Understanding how to work in the global context is a significant factor for project success. Prior global project experience impacted success in ten projects (with 18 outcome attributions). Lack of prior work experience with global projects can cause problems; as one manager indicated: "[the lack of success on the project] was a reflection of the fact that it was one of the first true global projects that my company tried to do ... it was different at the beginning, but we have learned a lot and now we are more successful at it." As another participant commented, lack of global work experience brings about a "sheer amount of virtualness." This in turn can lead organizational members to make erroneous assumptions and undermine their ability to deliver projects on time and on budget.

Attributions involving process variables

In this section we discuss how global boundary variables influence the use of coping process variables, and how these variables impact project outcomes in turn. Table III summarizes the number of attributions connecting global boundary variables with coping process variables. Note that "General" attributions were those that did not identify a specific subcategory of process variable. Table IV shows the number of attributions linking process variables to outcomes.

Coordination: communication. As Table III shows, communication was the type of process variable most affected by global boundary variables, especially by time separation and geographic distance. Interestingly, as Table IV shows, communication, however, was only the second most important coping process that affected outcomes. As we discuss in the next section, the adoption and use of task programming processes had a stronger impact on project success. Various forms of communications were used

Input variables	Communication			Sub total	Task programming				Cognitive				Sub total	Total unique attrib	
	General	Synch	Asynch		General	Project ctrls	Info tech	Other	Sub total	Shared belief	Shared knowl	Trust			Other
Global boundary variables															
Geographic distance	11	6	1	18	5	7	3	3	18	6	5	5	1	17	53
Time separation	11	8	3	22	7	6	2	5	20	2	1	1	2	6	48
Organizational boundaries	1			1	3	3		1	7	2	2			4	12
Functional differences	1			1		1			1		1			1	3
Cultural differences	10			10		1			1	5	2	1		8	19
Language differences	8	2	2	12		1		1	2		1		2	3	17
Other input variables															
Prior global experience	2			2		1			1			1		1	4
Prior experience w/team				0				1	1		2	4	1	7	8
Task context/dependencies	2			2	4	1			5		1			1	8
Total	46	16	6	68	19	21	5	11	56	15	15	12	6	48	172
Note: the numbers indicate the number of interviews in which the attribution was made															

Table III.
Attributions about the
effect of input variables
on process variables
(I → P)

Table IV.
Attributions about the
effect of process variables
on outcomes (P→O)

Process	General outcome	On budget	Costs and effort	Outcome Syst quality	On time	Meet reqs	Total
<i>Communication</i>							
General	15	1	1		3	1	21
Synchronous	9				1		10
Asynchronous	2		1				3
Subtotal	26	1	2	0	4	1	34
<i>Task programming</i>							
General	8	1	1		4		14
Project controls	13	1	1	2	1		18
Information technology	5					1	6
Other	6						6
Subtotal	32	2	2	2	5	1	44
<i>Cognitive</i>							
Shared beliefs	7			1	2	1	11
Shared knowledge	4		1	3	2	1	11
Trust	2						2
Other	2						2
Subtotal	15	0	1	4	4	2	26
Total	73	3	5	6	13	4	104

Note: the numbers indicate the number of interviews in which the attribution was made

in global IS projects. We found that many project teams used synchronous communication methods even if they often caused inconveniences such as adjustments of work hours or work days, or working late at home. While face-to-face communication was largely being replaced by advanced communication technologies such as teleconference or videoconference, many participants still indicated a preference for face-to-face communication. Face-to-face communication was especially effective when global project teams consisting of people from different cultures worked on critical, complex tasks. “We tried web meetings, conference calls, all those things and I can tell you there is no substitute for sitting down face to face. It allows us to build better working relationships with those people you can’t build over the telephone or via email. It allows us to get through more work in a short period of time,” said a participant.

Cultural and language differences affected the amount and frequency of communication, formality of communication, and people’s willingness to communicate difficult issues. A participant said, “Within our team, when people don’t understand things, just pick up the phone and call someone and ask questions. However, we don’t see that as much with our offshore partners. They tend to sit on problems and try to figure out those problems on their own rather than picking up the phone and calling us.” English proficiency greatly affected communication effectiveness. When people did not speak English fluently enough, sharing complex ideas became very challenging, if not impossible.

While advanced communication technologies have diminished the impact of geographic distance on team communication, time separation continues to be a

problem requiring special processes. To cope with this, teams implemented daily or weekly meetings, regular conference calls, and regular reporting of the project. Some teams opened “a [communication] command center structure that runs 24 hours” to “take the time difference out of the picture,” as one participant described.

Some participants also stated that they tried to make communications as clear as possible and free of ambiguities, which required more formal communication practices and more detailed documentation. For example, one team implemented “understanding documents” for members to record their understanding of what they had been told by other members to avoid hidden issues and misunderstandings. Some teams also rotated people to different locations and encouraged frequent on-site meetings. By so doing, project team members were able to overcome some of the problems due to cultural differences, time separation and geographic distance.

Coordination: task programming. While the use of task programming mechanisms is effective for routine tasks, they are also key success factors in global collaboration in which communication is impaired. “Basically in the global project, you need to get in place lots of things that you can take for granted when you are localized,” said one participant. As we discussed in the previous section, while communication is important for global IS project success, task programming is most effective in offsetting the communication problems in global teams. This is an interesting finding because the literature on team communication argues strongly about the importance of communication for teamwork and coordination (Fussell and Krauss, 1992; Kiesler and Cummings, 2002), especially with globally distributed teams (Cramton, 2001; Kiesler and Cummings, 2002), but our study shows that when global boundary variables make it difficult to communicate, teams adopt alternative coordination mechanisms like task programming.

Among the most frequently mentioned task programming mechanisms were rigorous documentation, common processes, strict project controls and detailed project planning. Effective global collaborators put more effort and rigor into formal documents and processes, as illustrated by these comments made by two participants: “when you work a distance from a client and with a geographically dispersed team, you have to have clearly written requirements” and “the fact that the programmer is not seated next to you makes you have to write specifications in much more detail.” Common processes helped eliminate a significant amount of communication by establishing protocols and clarifying issues related to processes and tools so that teams could channel their attention to more substantive issues and problems. For example, one organization had developed a common global solutions management system across all development sites.

Other task programming processes used for coordination include task organization, division of labor, collaboration technology tools and the adoption of common technical environments at each site. Task organization involved placing key personnel at other sites, having redundant roles and coordinator roles, and assigning tasks based on local expertise and workflow needs. Division of labor not only helped de-couple dependencies across sites, but also helped members locate expertise, allocate tasks more effectively, and work more independently, thus reducing the need to communicate. Project managers also advocated using a portfolio of collaboration technologies to offset their reduced communication including project automation tools,

instant messaging, tele and video conferencing, web meetings with shared document capabilities. Teleconferencing was most often cited as an effective technology-mediated communication method, while e-mail was considered inferior when common understanding was needed.

Time separation was the most frequent reason for the use of task programming mechanisms (20 unique attributions). Teams separated by many time zones required more task organization and coordinated meeting and communications schedules. As two participants stated: "one of the things we did was ... to have a slight overlap of working hours so we managed to set up meetings that were sort of at the end of someone's day or the start of someone else's day ..." and "what we do is make sure that whatever deliverables we are sending people offshore is with them before they reach the office ... so they don't lose any time essentially in the process."

External organizational boundaries had significant impacts on process variables such as project control mechanisms in three cases. For example, client companies needed tighter control mechanisms to prevent the outsourcing partners from opportunistic behavior. "They [outsourcing partners] don't plan; an employee is sick or basically needs to take a vacation but he's in the middle of your project," said one participant. Multiple companies also need to negotiate their work procedures, processes, and quality standards in the early stages of their project lifecycle.

Team cognition. Geographic distance was discussed as the most significant factor affecting members' ability to develop shared knowledge (five participants). Ironically, it is precisely in geographically dispersed contexts in which communication is impaired that shared knowledge can be most valuable for coordination. As one participant commented "knowledge sharing is important and it is imperative for programmers there to understand the business ... we traveled to India to educate them about the specifications." Knowledge sharing also has positive interaction effects with communication and task programming because it provides a common knowledge base that can make communication more effective and the use of task programming mechanisms better tailored to the needs of the task.

Shared beliefs and shared knowledge had equally strong effects on project outcomes (11 unique attributions). Shared beliefs included things like common goals, shared vision about the project, thinking like "one team" and a common understanding of processes and issues. Such shared beliefs helped members work more effectively with less communication and made their communication more effective because they had more common ground and shared vocabularies. Developing shared beliefs was most negatively affected by geographic distance (six participants) and cultural differences (five participants).

Participants discussed the difficulties of establishing trust in the presence of global boundary variables. As one participant remarked, "there can be a certain amount of reluctance to get involved with global projects because they can see the people there [offshore] as a threat to their own jobs, but we have found that people who have actually got involved with offshore projects and got them to work are quite positive about them." Only two participants indicated the importance of trust to project outcomes. But those who discussed trust felt strongly about its importance. As one participant noted: "if you don't trust your team there is no point in collaborating. You have to trust your offshore team to give you their best results."

Additional data collection and findings

While most projects (21 out of 22) were reported to be successful, some participants hinted at critical problems in early stages of the project. To gain more insights into our findings, we collected additional data using a web-based online survey instrument. This online survey asked respondents to indicate whether or not their project was failing at some point in time in terms of meeting user requirements, on-time execution, within-budget execution, and team satisfaction (using separate questions for each). In cases where their project was failing at some point in time, the respondents were also asked to provide the reasons for failure in that particular success measure and then describe corrective processes that were employed to get their projects back on track.

We solicited responses from the 22 project managers in our original study sample, but six managers could not be reached. We received 13 responses from the remaining managers, but one response was incomplete, yielding an effective response rate of 75 percent (12/16). While four out of 12 respondents reported that their projects were successful throughout the entire project, the other eight respondents reported that their projects were failing at some point in time. The reasons for the temporary failures varied. But, the most frequently mentioned reasons were: flawed, unrealistic, inaccurate estimation of time and cost; and vague, ill-defined, ever-changing user requirements. While these problems may also be common to domestic software projects, these problems tended to be exacerbated in global software projects due to the increased complexity of task interdependencies among multiple sites and vendors.

We found that global software teams overcame the temporary failures by adopting effective corrective task processes. Most importantly, global software teams engaged users, senior management, and vendors in increased communication, which included discussions regarding the status of the project and redefinition of user requirements, project duration, and project budget, among other things. Improved communication among project stakeholders and stronger project governance helped the projects get back on track. Another important correcting process was restructuring the project team. For example, project managers were replaced with seasoned managers with global project experience, and key users were pulled out from business units to join the project team.

In eight cases, global software teams implemented improved project processes to turn around their failing projects. They revised quality assurance processes to meet the required quality, implemented rigorous task monitoring procedures and robust cost management procedures, and increased project management discipline. Finally, a common fast fix adopted to correct temporary failures was to simply revise the project schedule and budget. However, this simple solution was not an option in many cases because global software projects had firm deadlines and budgets that could not be changed.

Discussion

This study found that in IS projects with multiple global boundaries, time separation was by far the most significant barrier, followed by cultural differences, whereas distance alone did not appear to be a major obstacle. This finding suggests that it is not so much face-to-face interaction that is vital to global IS project success, but rather the ability to interact synchronously and interactively when needed, which is not affected

by distance, but severely hampered by time separation. This is particularly true for more complex projects with highly interdependent tasks, which require the initiation of more communication incidents (Maznevski and Chudoba, 2000). Time separation is a special global team boundary variable because it can affect project performance both positively and negatively. When activities can be well coordinated and synchronized, time separation can be used advantageously. Generally speaking, time separation can work well when tasks are well defined and when task programming mechanisms can provide effective coordination. On the other hand, time separation becomes a substantial collaboration barrier when the tasks are less well defined and more uncertain, thus requiring frequent interactive communication. While distance is still a barrier, it was no longer considered to be a major factor by many of our participants. People have learned to handle geographic distance from their experience with projects involving multiple locations, relying extensively on suitable collaboration technologies. This study suggests that in most cases, the problems involving geographic distance often correlate with the presence of other global boundary variables (e.g. cultural differences, time separation, etc.). Once all these other boundaries are accounted for, the effects of pure geographic distance diminish substantially.

It is interesting to note that despite the risks that result from multiple global boundaries and despite the fact that one-third of the projects were failing at some point, 21 of the 22 projects were deemed successful in the end. This surprising finding suggests that the aggressive push to meet corporate expectations inspires project managers to implement effective coping processes that overcome global barriers at some point during the project, resulting in ultimate project success. These coping processes involved task programming, communication and team cognition. Specifically, project teams made special efforts during the initiation phase to create a common development environment and common understanding as well to organize development tasks in order to reduce dependencies across development sites. While some projects implemented these coping processes early on and were successful throughout the duration of the project, others were not so successful at the beginning, but later made adjustments and implemented some of these strategies to save failing projects. Those adjustments included processes that facilitated communication –, e.g. more frequent communication, more flexible work schedules to increase work overlap hours – and processes that reduced the need to communicate synchronously –, e.g. more detailed documentation, more effective use of task programming mechanisms and tighter project controls.

Coordination through task programming was the most frequently discussed process adopted to overcome the barriers imposed by global boundaries. Task programming mechanisms are generally more useful when tasks are certain, unambiguous, structured, and routine. But they are also used quite effectively in distance-separated contexts to minimize task dependencies, reduce the need to communicate, and increase accountability. Communication was the second most frequently discussed coordination mechanism employed to bridge global boundary variables. Communication is particularly effective when tasks are uncertain, ambiguous, unstructured, and less routine. Because communication is hampered by global boundaries, teams employed special communication approaches in the early stages of their projects, such as:

- communicating more frequently or regularly;
- communicating as clear as possible and employing more detailed documentation and formal communication processes; and
- establishing a common platform for global collaboration, which included common procedures, tools, and technologies to reduce the need for global communication.

While technology-mediated communication is now commonplace, many participants still felt that there is no substitute for face-to-face communication. Early face-to-face meetings were especially effective when global project teams consisting of people from different cultures were working on critical, complex tasks.

This study has confirmed the two types of processes associated with the I-P-O framework we presented in Figures 2 and 3. The first are the traditional task processes that teams use to get the work done, identified by Powell and colleagues (2004). These processes are often negatively impacted by global boundary variables. However, effective teams implement special coordination and cognitive processes (e.g. regular meetings, frequent travel, detailed specifications, developing shared knowledge early, etc.) to overcome potential deficiencies in the traditional task processes when bridging global barriers.

Organizations often experience problems in global IS projects because of their lack of awareness about the effects of global boundaries, particularly when managers have little experience with global projects. Once managers become aware of the impact of these boundaries, they implement special processes or mechanisms to bridge them. These processes are geared towards raising awareness about several task-related issues, including cultural diversity awareness, presence awareness (i.e. who is where and when), task awareness (i.e. what has been done and when), and local context awareness.

Global barriers are often overcome with coping processes suited to the task context, but with additional costs, effort and stress. For example, in time separated contexts with little or no working overlap, teams may implement a 24 hour-a-day availability policy. Teams also implement other tactics like: frequent travel; relocation; communication and collaboration tools; extreme attention to detail in documents, requirements, and specifications. All of these additional measures come at additional financial cost and burden on personal lives. The hidden costs associated with the additional time and effort required to work across global boundaries reduce the attractiveness of global IS projects. Given the current trends in offshore outsourcing, we were surprised that we did not hear from many project managers that global collaboration arrangements were established primarily to reduce costs. Quite the contrary, our findings suggest that global collaboration, regardless of outsourcing and off-shoring arrangements, adds costs that are not always captured in the project budget and contract.

Limitations

Our study has limitations. First, the sample is relatively small, consisting of 22 interviews. Nevertheless, we managed to achieve theoretical saturation in the text data (i.e. the last few interviews did not contribute new insights) and obtain very rich accounts about the effects of various global boundary variables and about the

processes that are most effective in overcoming these global barriers. Second, our study is limited by inherent problems of interview research, such as lack of statistical validation and interpretation bias. Further quantitative studies are needed to develop a better understanding of the factors leading to global IS project success. Finally, our findings may be limited to the idiosyncrasies of the companies that were studied. This limitation potentially weakens the generalizability of our findings to a broader population of organizations. However, although our sample did not represent all industries and organizations, it included interviews with project managers from seven different companies, covering a variety of IT settings.

Conclusions

Despite these limitations, this study makes an important contribution to the literature on global IS project teams because it is the first study to investigate the relative effects of multiple global boundaries and other input variables on global IS project outcomes and how different processes mitigate these effects. Time separation and cultural differences were found to have the most significant impacts on global project outcomes. Working in global contexts is a difficult undertaking and thus IS projects have the potential to fail. Despite multiple global barriers, many projects have succeeded by employing task programming, communication and cognitive processes that are custom-fit to alleviate the problems and challenges presented by these barriers. Future research is needed to quantitatively validate our findings and further investigate the complex inner workings of global IS projects.

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Appendix

Interview instrument for global IS project success study

General Questions (Input Variables → Outcomes)

- How do you measure success/failure in a global IS project?
- Generally speaking, which global barriers/boundaries had the most significant effect on project performance? And, how did those factors affect various dimensions (e.g., time, cost, quality, user satisfaction) of project performance?

Project Specific Questions (Input Variables (Process; Process (Outcomes)

- Was the project completed on time?
- Was the project completed on budget?
- Did the system meet requirements/quality standards?
- Were users/team members satisfied?
- How successful was this project? Why? Please explain
- What processes/mechanisms/strategies/methods/technologies were used to cope with these situational factors? And how did they affect various dimensions (e.g., time, cost, quality, user satisfaction) of project performance?
- If you do the project again, how would you do it differently?
- What makes global projects different from domestic projects?

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